

7. Project Documentation Phase

1. INTRODUCTION

The Project Documentation Phase ensures that **ComicCraft** is fully understandable, maintainable, and operable by external developers. This stage establishes detailed setup instructions, dependency manifests, environment configurations, API specifications, and usage guides necessary for seamless onboarding, execution, and contribution.

2. OBJECTIVES

Phase 7 focuses on creating clear technical artifacts and execution guides. Key objectives include:

- Provide a comprehensive high-level project overview and feature breakdown.
- Specify core system prerequisites and execution environment standards.
- Document step-by-step setup and local installation procedures.
- Define environment variables and secure credential management patterns.
- Detail repository folder organization and component boundaries.
- Publish RESTful API endpoint specifications and schema definitions.
- Provide an end-to-end user guide for comic generation workflows.
- Document testing instructions and quality assurance command steps.
- Identify project limitations, known constraints, and future enhancement roadmaps.

3. PHASE WORKFLOW

The documentation framework spans setup instructions, dependency definitions, local server operations, interactive API endpoints, user interface workflows, and complete comic-generation operational pipelines.

4. DETAILED EXPLANATION

This phase consolidates technical requirements and startup commands needed to deploy ComicCraft locally. To preserve security, external API credentials and sensitive keys must be injected strictly via environment variables (such as `.env` configurations) and barred from version control using strict `.gitignore` rules.

5. GITHUB-READY CODE / STRUCTURE

Below are the foundational dependency definition and environment template files for the repository:

```
# requirements.txt
fastapi
uvicorn
jinja2
python-multipart
google-generativeai
diffusers
transformers
fpdf
Pillow
accelerate

# .env.example
GEMINI_API_KEY=your_gemini_api_key_here
HF_API_KEY=your_huggingface_api_key_here
```

6. GITHUB FILES & DELIVERABLES

Key Technical Documentation Artifacts:

- `README.md` — Master repository guide, project overview, and quick-start instructions.
- `docs/installation.md` — Detailed setup guide, dependency installation, and local execution steps.
- `docs/api.md` — Detailed API routes, input payloads, and response schema documentation.
- `docs/user-guide.md` — Visual walkthrough for interacting with frontend forms and exporting PDF comics.
- `.env.example` — Standardized template for required environment variable configurations.
- `.gitignore` — Rule set to exclude virtual environments, generated PDFs, and secret key files.

7. EXPECTED OUTCOME

By the conclusion of Phase 7, the entire operational and developer ecosystem of **ComicCraft** is thoroughly documented, configured, and verified. Actual project completion relies on the presence and completeness of these markdown guides and configuration files in the student's repository.

8. PHASE CONCLUSION

Phase 7 provides a structured foundation for the final presentation and deployment stages of **ComicCraft**, ensuring the repository remains organized, reproducible, and ready for GitHub-based evaluation and peer review.